

Arohi

An interface to connect disaster-affected individuals with rescue teams in real time, enabling faster decisions, clearer coordination, and efficient delivery of aid.



01 Problem Area

When a disaster occurs, it is usually managed by the local government, and how effectively it is handled largely depends on the level of preparedness in place. However, the ones who suffer the most are the people trapped in the disaster and the rescue teams on the ground — often due to insufficient information and unclear communication.

In the face of disaster, it's crucial to have a unified platform that brings people together and enables them to support one another.

Bird's-eye View

"The people stuck in disaster have no idea who to contact where to contact"

"The services have no point of contact and understanding of real situation"

02 Situation Mapping

When a disaster occurs, the way emergency services are contacted can be mapped to understand the flow of information. Although disaster situations are complex and overwhelming, mapping the process makes them more comprehensible helping us see how different services are involved, how information moves between them, and at which stage each stakeholder intervenes.



Identified services involved :

1. 112 emergency services
2. Local police
3. Medical team and hospitals
4. Media and volunteer
5. Local civics board
6. State disaster management team
7. National disaster management team

- Understanding how information flows and what types of information flow
- Identifying key hotspots for information transfer
- Understanding who the main players are and the priority in which work is carried out

Identified data points :

1. Count of death and injured
2. Affected people
3. Type of disaster
4. Scale of disaster
5. Cause of disaster
6. Affected area
7. Result of disaster
8. Time of occurrence
9. Location and affected span of area
10. Result of disaster

- Required by first hand reporters
- Required to further send for help
- Required to control the situation
- Required to for media/ situation debrief analysis

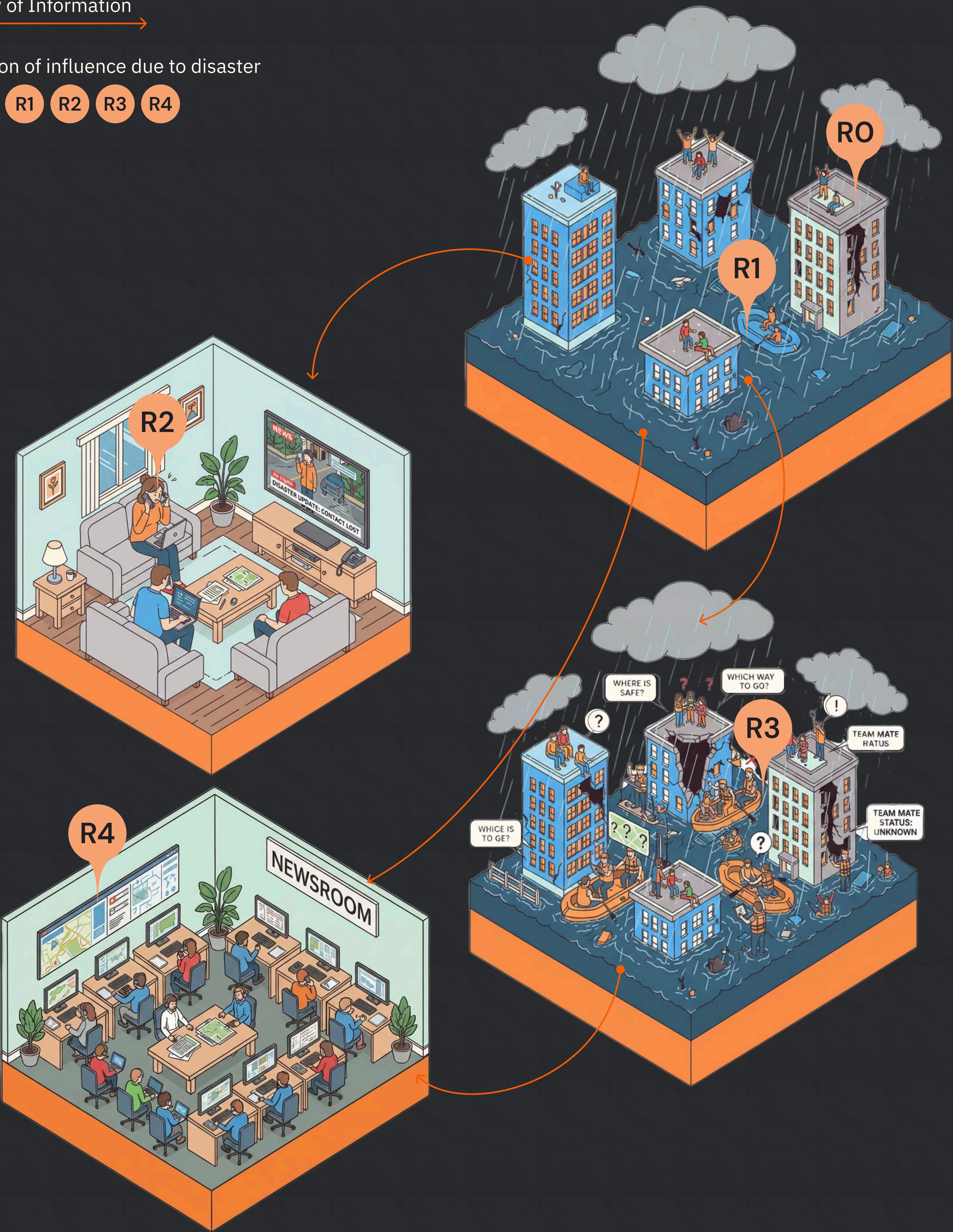
03 Ecosystem Mapping

When a disaster occurs, the way emergency services are contacted can be mapped to understand the flow of information. Although disaster situations are complex and overwhelming, mapping the process makes them more comprehensible helping us see how different services are involved, how information moves between them, and at which stage each stakeholder intervenes.

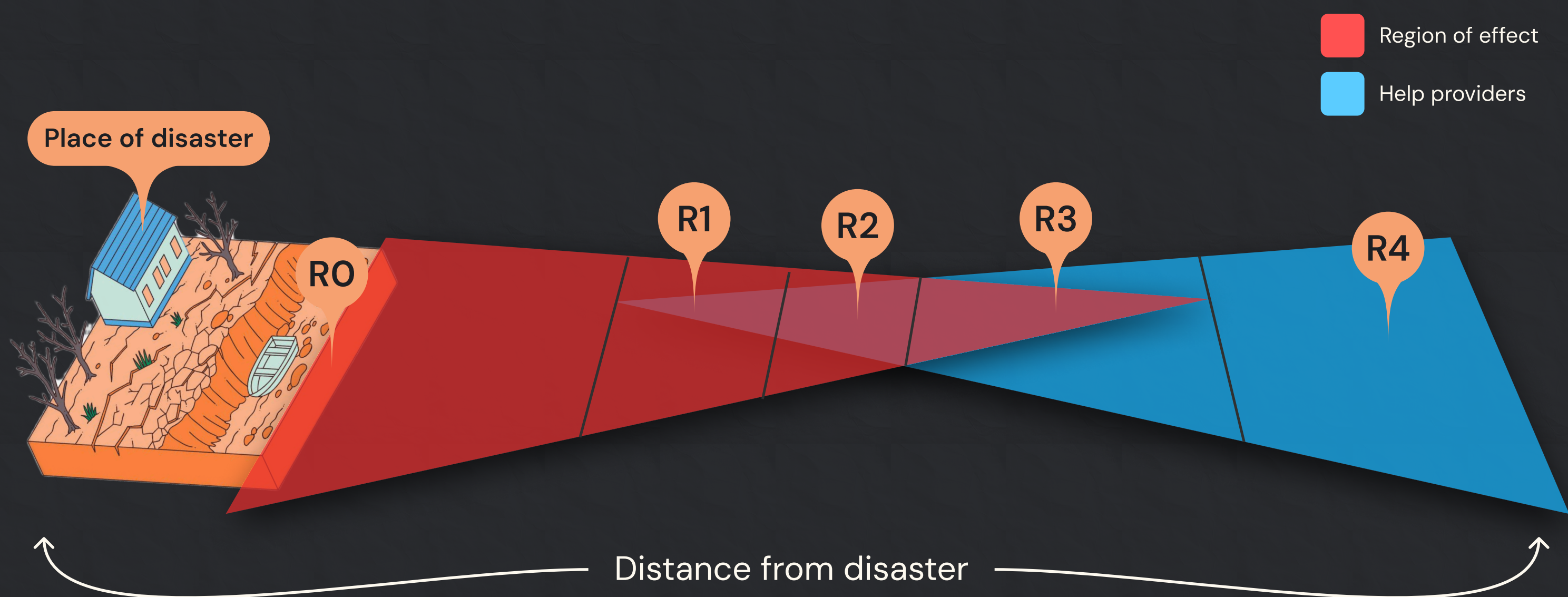
Flow of Information

Region of influence due to disaster

- R0
- R1
- R2
- R3
- R4



04 User Pain-points and Expectations



R0: People stuck in disaster			
Pains	- Physical pain - Confusion	Expectations	- Medical help - Status of help - Ability to call for help
R1: People who are stuck due to disaster			
Pains	- No one to contact - Confusion regarding situation - Save themselves and their family	Expectations	- Situation awareness - Ability to call help - Safety evacuation procedures
R2: Families of people who are stuck (R0,R1)			
Pains	- Needs reassurances - Confusion regarding situation - Anxiety about contacting their family	Expectations	- Contact info for their family - Status and location of their family
R3: Disaster Management Team, Emergency services etc.			
Pains	- Don't know where and how to contact - Unable to figure what kind of help required - Which information to trust - How to reach onsite - Managing all requests at once	Expectations	- Status of situation - Who to contact onsite - Area wise division for red, orange and yellow regions - Contact conveniently for more help required - Situation control - Updating status to higher authority effectively
R4: Government, Media			
Pains	- Late status update - No idea how to help and which service to provide - How/whom to send fund money	Expectations	- Latest situation update - Status of help provided - Causalities and their family member contacts

05 User Personas with Tasks

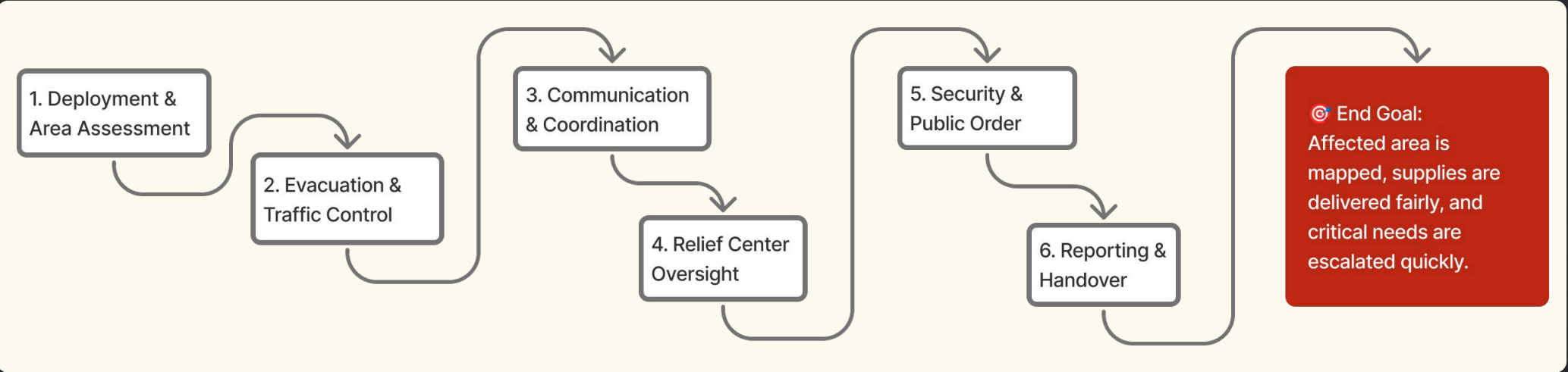


Name: Sub-Inspector Arvind Patil
Age: 42
Location: Sangli, Maharashtra

- Behaviors:**
- Communicates via walkie-talkie when mobile fails
 - Tags households with chalk/paint to indicate “visited” or “needs help”
 - Stays in contact with the command center every 3–4 hours

- Pain Points:**
- Unreliable mobile networks and overloaded police radio channels
 - Physical fatigue and limited access to food, dry clothing, and rest
 - Crowd panic, misinformation, and resistance to evacuation
 - Lack of up-to-date area maps or tech tools to monitor affected zones

- Goals:**
- Ensure orderly evacuation of residents from flooded areas
 - Coordinate effectively with fire and rescue teams, local leaders, and NGOs
 - Maintain law and order at evacuation and relief points
 - Respond to distress calls from stranded citizens
 - Monitor and update status reports to higher authorities



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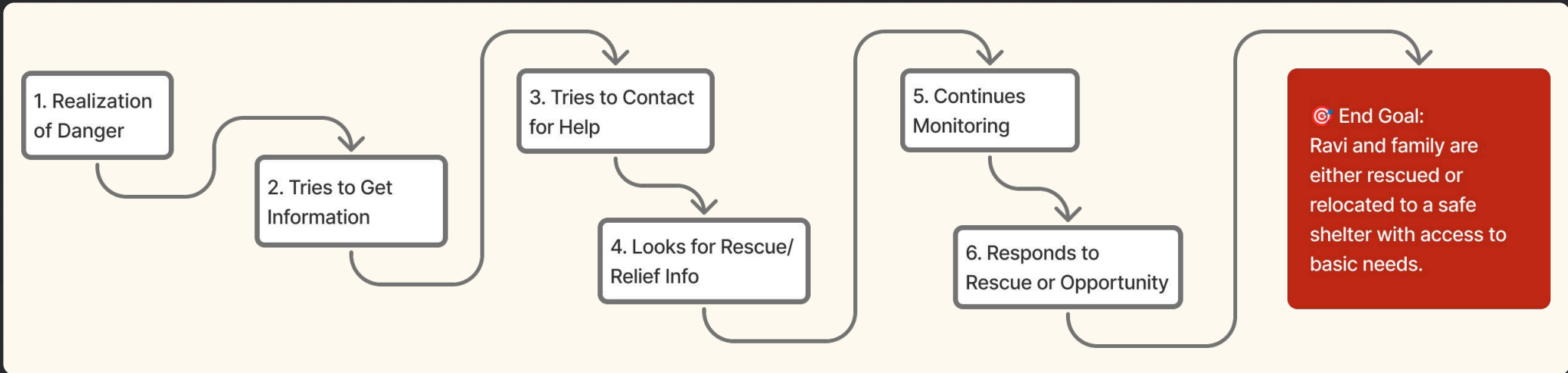
Name: Ravi Deshmukh
Age: 38
Location: Nashik, Maharashtra, India

- Behaviors:**
- Tries calling emergency helplines
 - Keeps checking weather apps, though updates are delayed
 - Listens to radio on an old battery-powered device if available
 - Relies on neighbors for updates and help

2

- Pain Points:**
- Unclear or conflicting information from news and government sources
 - Mobile phone signal drops frequently; unable to access data-heavy services
 - No power for charging device
 - No clear way to signal location for rescue

- Goals:**
- Ensure the safety of his children, mother, and wife
 - Find a way to communicate with authorities or local help groups
 - Preserve mobile battery life for critical communication

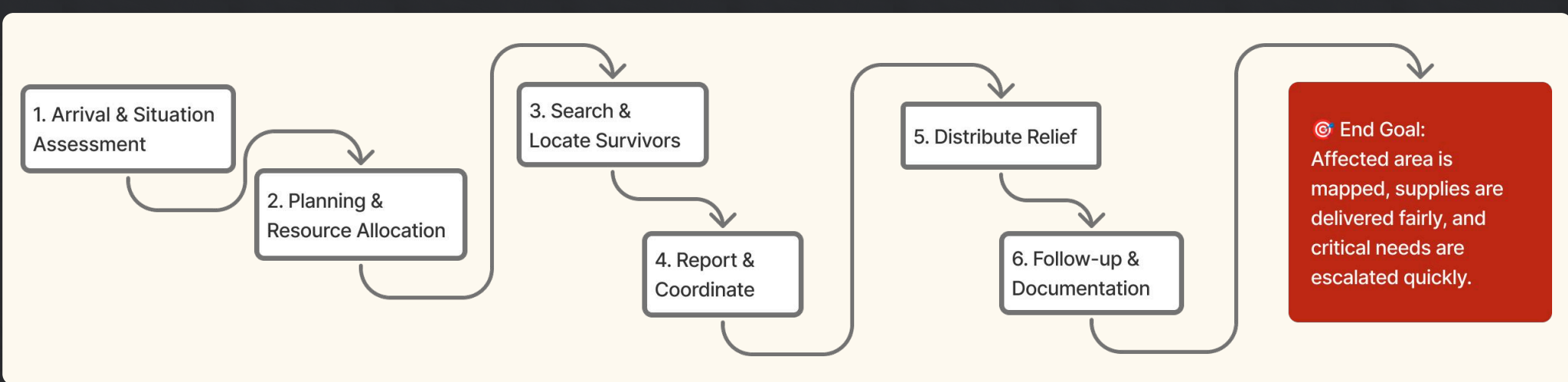


Name: Sanvi Saranjame
Age: 32
Location: Nashik, Maharashtra, India

- Behaviors:**
- Communicates via walkie-talkie when mobile fails
 - Tags households with chalk/paint to indicate “visited” or “needs help”
 - Stays in contact with the command center every 3–4 hours

- Pain Points:**
- Disorganized local data—no clear maps or lists of affected households
 - Battery drain from using communication and mapping tools constantly
 - Poor coordination between government and NGO efforts

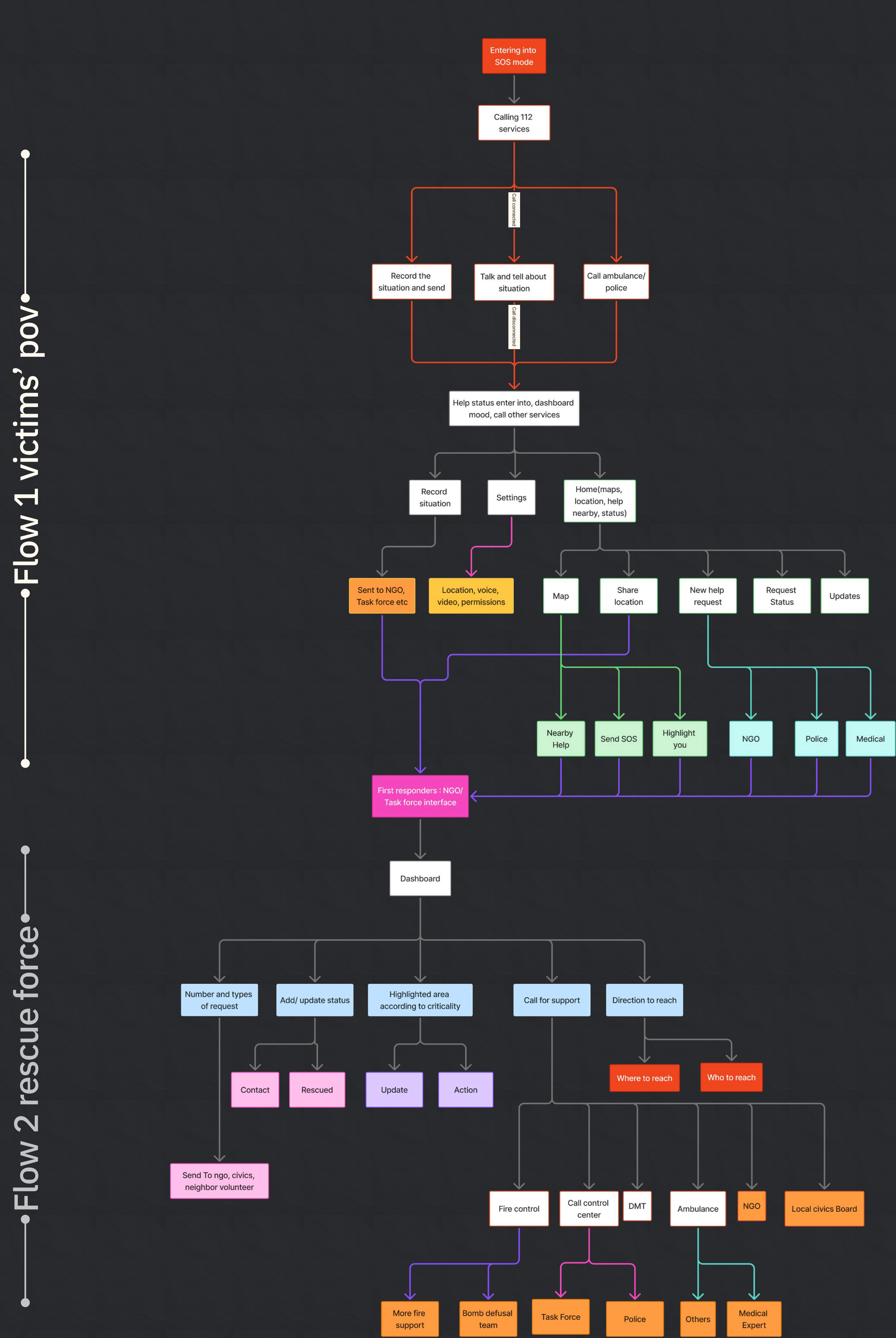
- Goals:**
- Reach as many stranded people as possible in minimal time
 - Coordinate with other teams and rescue boats
 - Keep herself and her team safe
 - Ensure fair and transparent distribution of resources



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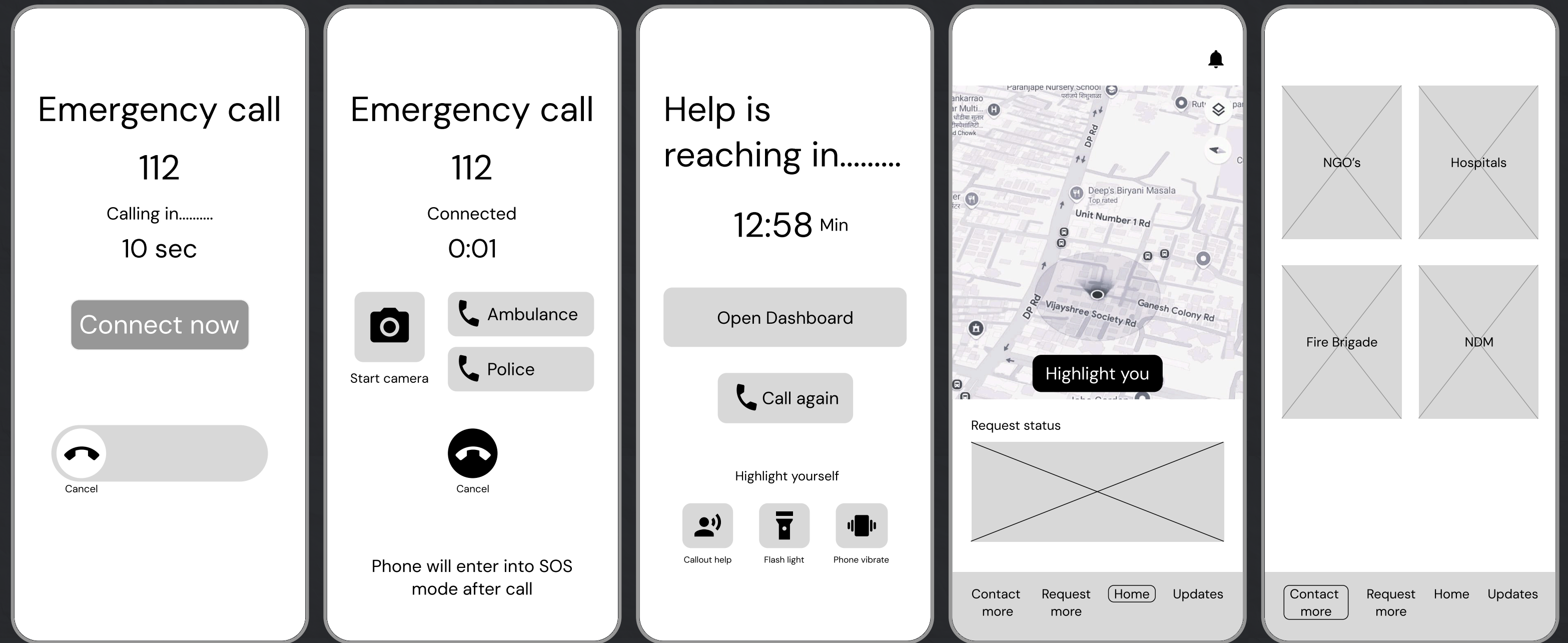
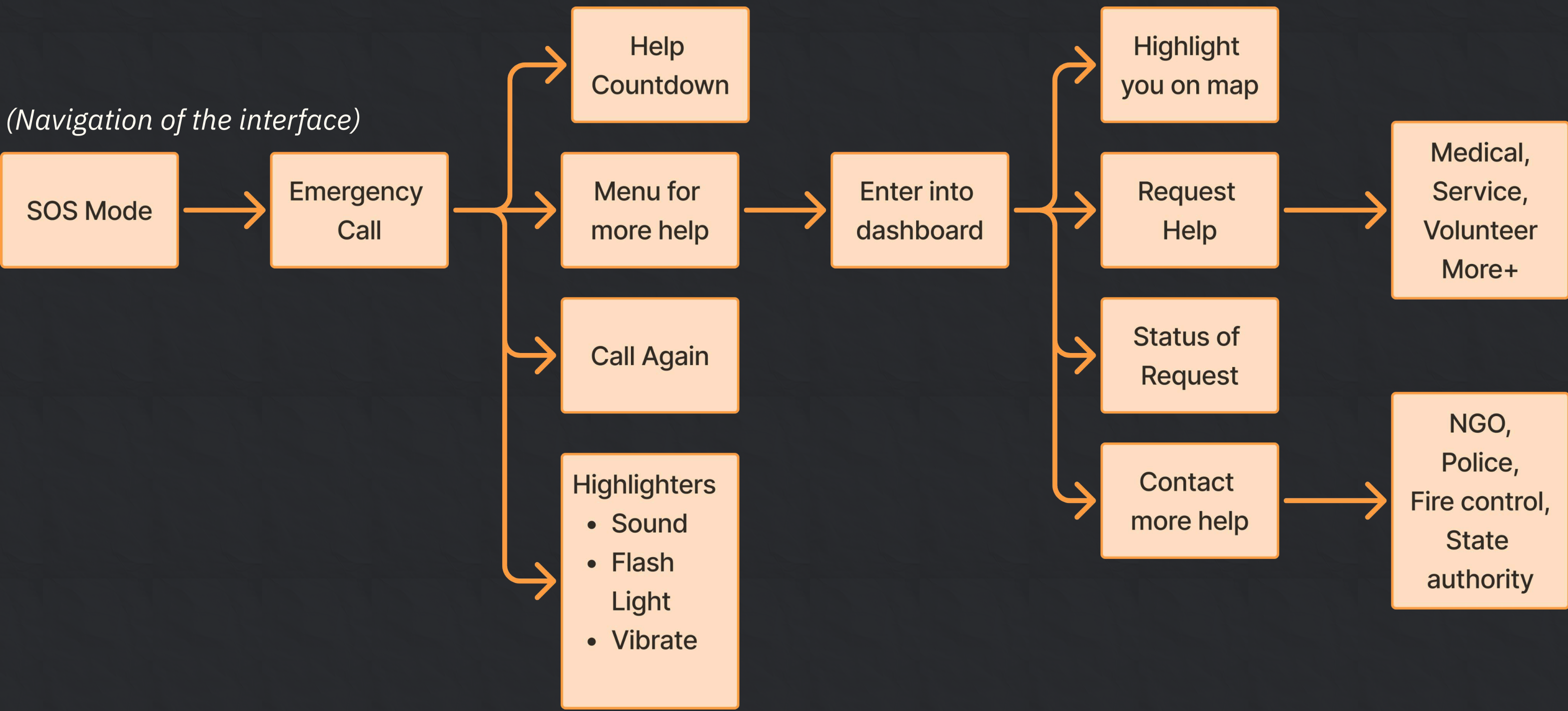
06 Information Architecture

From the key areas where we identified potential data loss and analyzed the tasks performed by each user in different regions, the following information architecture was developed. It prioritizes features to ensure maximum support — both for rescue forces searching for critical information and for individuals affected during a disaster.

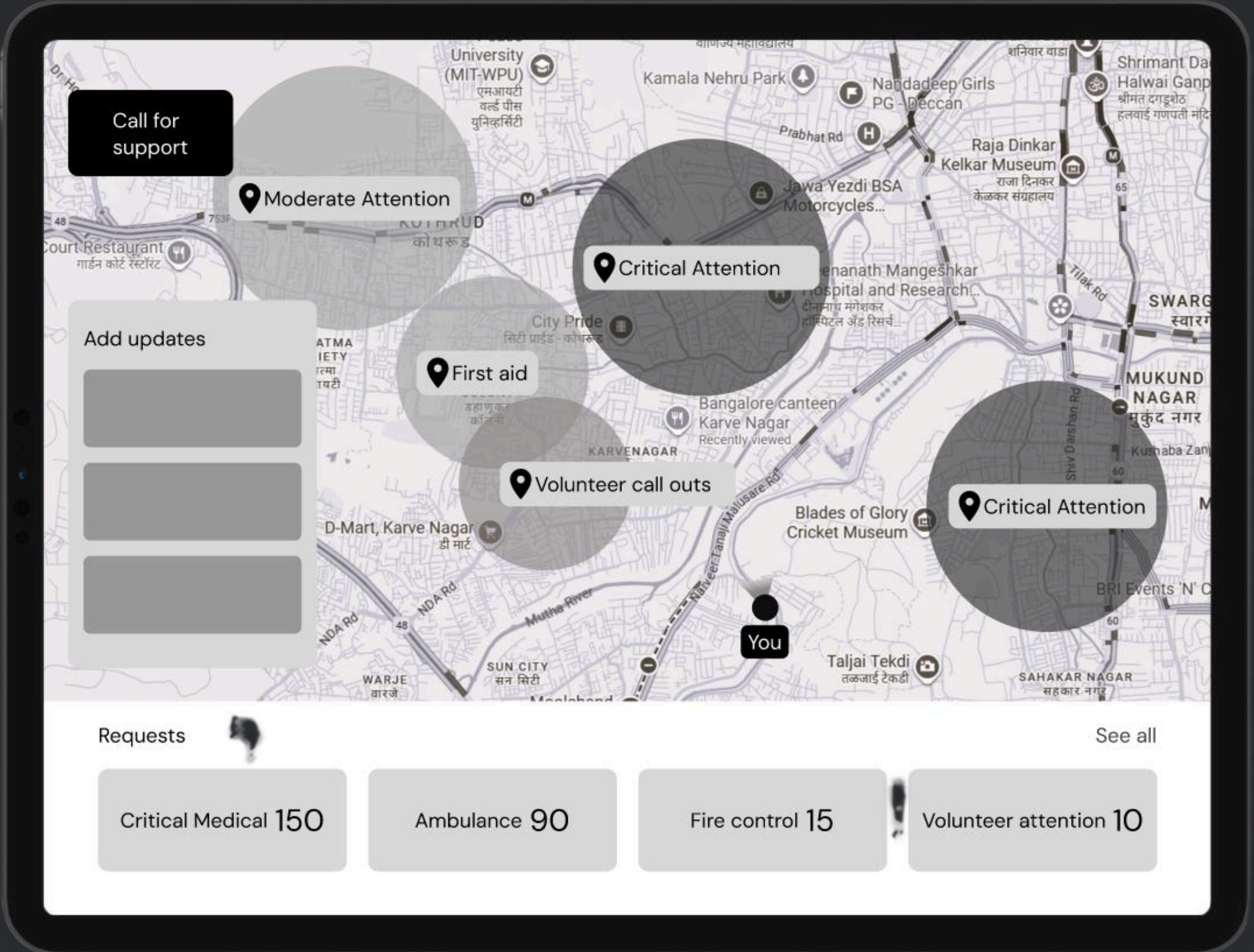
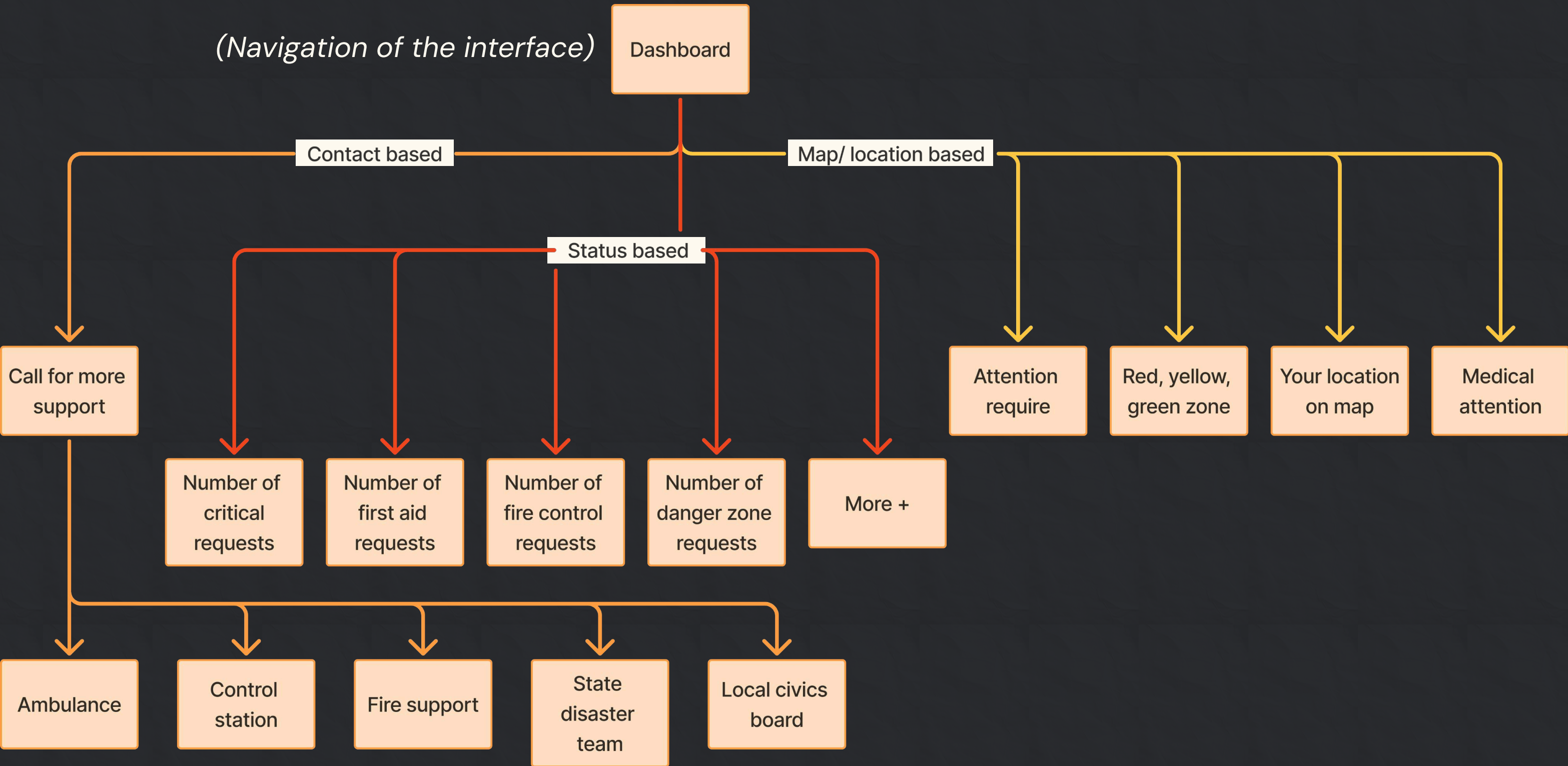


07 Wire-Frames and Navigation

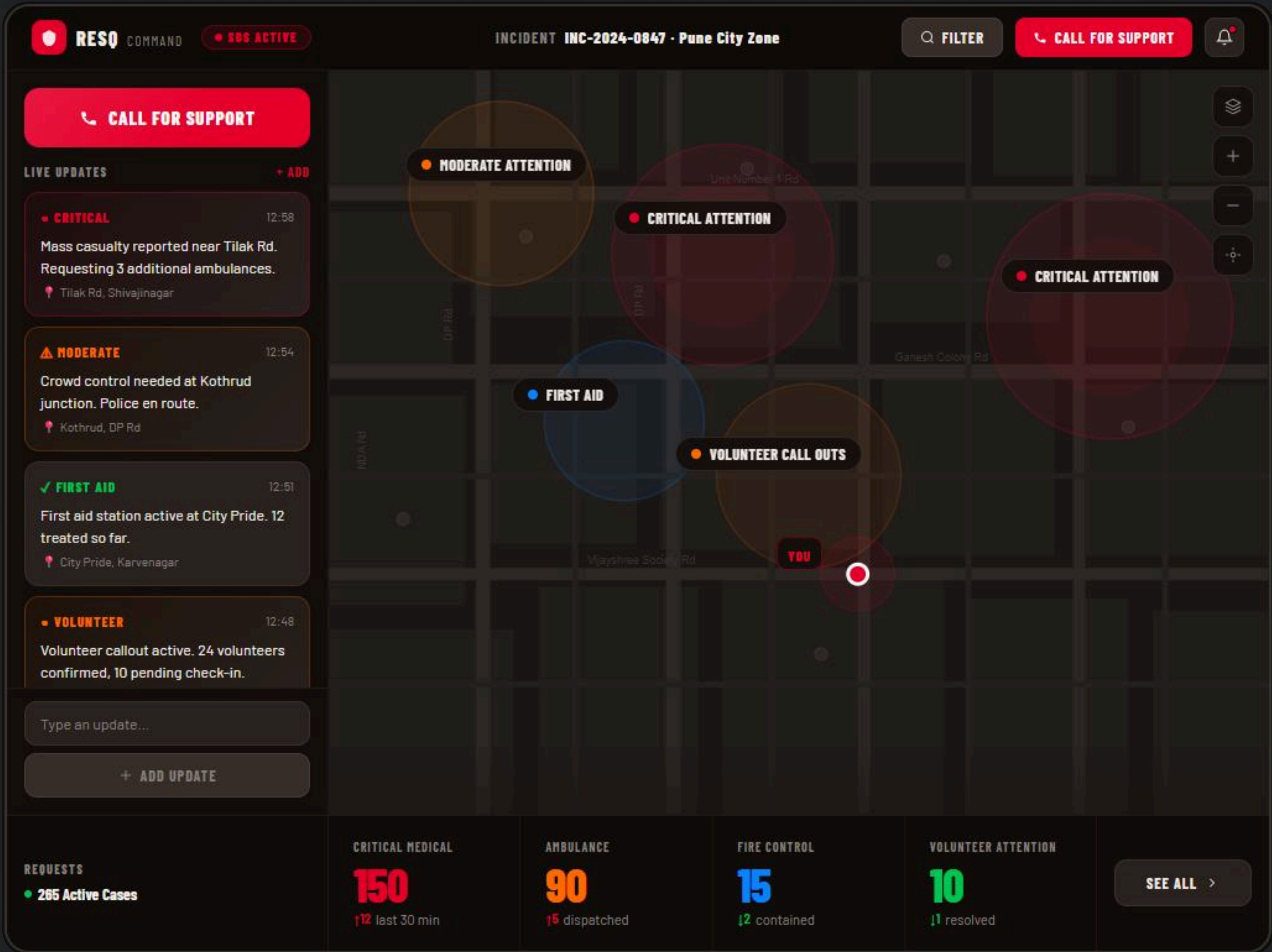
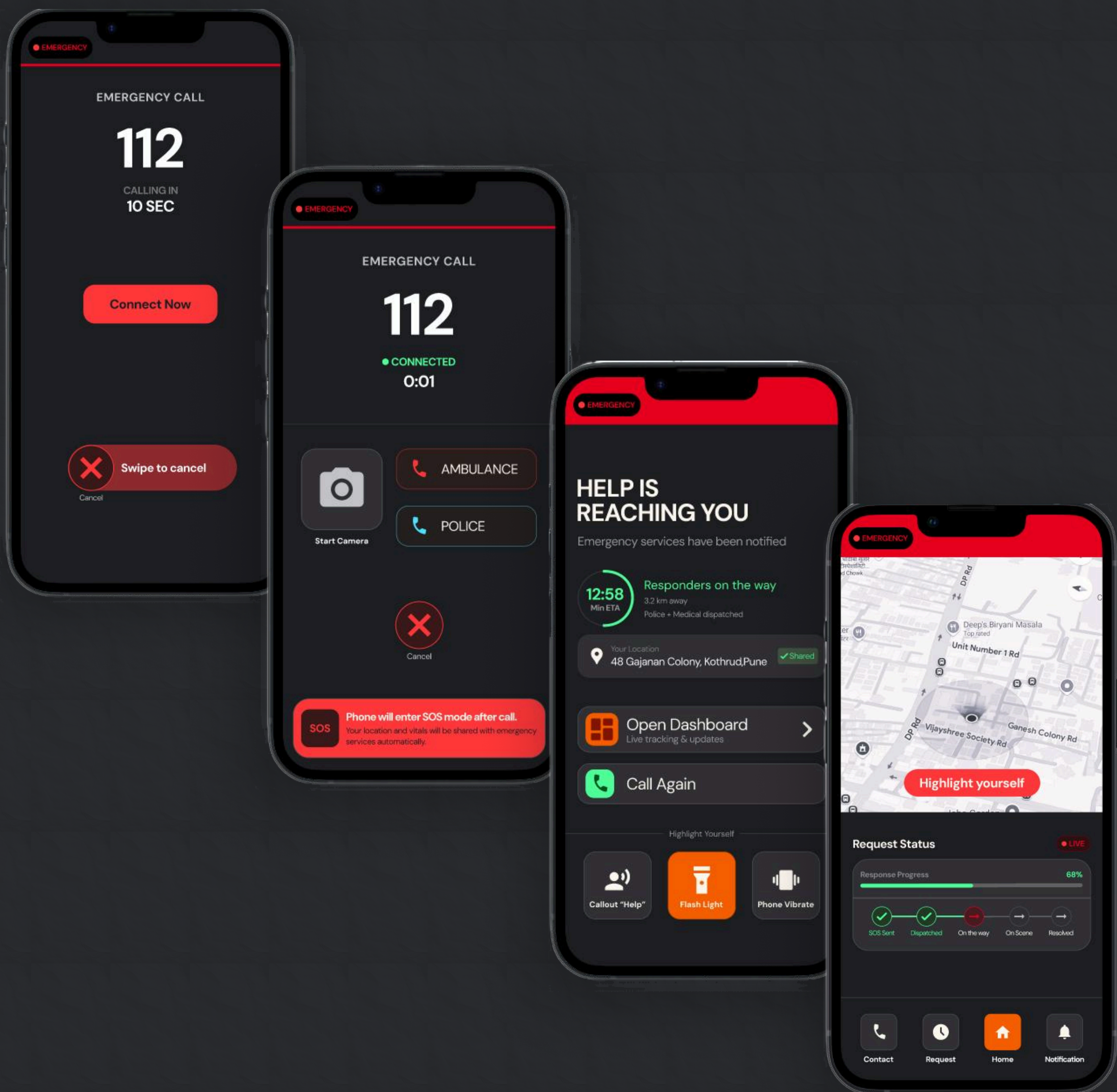
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• Flow 2 rescue force



08 UI Screens



10 Conclusion

Designing UI screens is relatively simple, but this solution depends heavily on a stable network — something that is often unavailable during disasters. Therefore, as technology evolves and more resilient communication systems are developed, the interface will also need to adapt. Depending on the new device and infrastructure, the future UI may look and function differently, as envisioned below.



However, one element that will remain constant is the information architecture and navigation flow, as this is where UX plays its most critical role during a disaster.

11 Links and References

1. Detailed report
2. Research Papers

09 Prototype